

# Station 1: Studio lights

*Build three-point lighting one source at a time, shoot it with and without fill, then walk the key light around the clock and capture every hour.*

**Where your shots go:** On the station computer, the capture page shows the studio camera's feed. When you like the shot, press **Capture** (or the space bar); the still goes to the wall. Name nothing; just press it. Fill / no fill, then the two clock series.

## Terms to master

| Term                       | In one line                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Key light</b>           | The main light on a subject, placed to one side and above eye level so the face has shape.                                                                                     |
| <b>Fill light</b>          | A second, weaker source or a white card on the opposite side from the key, controlling how dark the shadows get.                                                               |
| <b>Back light</b>          | A light behind and above the subject, aimed at the hair and shoulders, that separates them from the background.                                                                |
| <b>Kicker</b>              | A light from behind the subject at an angle, raking the cheek or jaw with a thin bright edge. One of the names for the third point.                                            |
| <b>Hard and soft light</b> | Hard light comes from a small source and makes crisp shadow edges; soft light comes from a big source and makes gradual ones. Size relative to the subject is what decides it. |
| <b>Catchlight</b>          | The small reflection of the light source in a subject's eyes. It gives away the source's shape, size, and position.                                                            |
| <b>Color temperature</b>   | How warm or cool a light is, in kelvin: 3200 K is tungsten orange, 5600 K is daylight blue-white.                                                                              |
| <b>Reflector</b>           | A white, silver, or gold surface that bounces the key light back into the shadow side of a face. Fill without a second light.                                                  |
| <b>RGB light</b>           | An LED light that can be set to any color, not just warm or cool white. For colored back lights, colored walls, and color contrast between the points.                         |
| <b>Ring light</b>          | A circle of LEDs around the lens. Even, shadowless light from the camera's own position, with a single round catchlight in each eye.                                           |

Gear at this station: **large and small bi-color LED panels, reflectors** (white and silver, plus foam core), **adjustable RGB lights** that can be any color, a **ring light**, a stool, and the studio camera, whose feed is on the station computer's capture page.

One rule covers most of it: **the bigger the light looks from where the subject sits, the softer it is**. A large panel is softer than a small one; any panel gets softer as you bring it closer. Small and far away is hard light with crisp shadow edges. There is no diffuser at this station; size and distance are the two dials.

Positions use a clock face seen from above: the subject sits at the center facing the camera at 12.

## Recipe

1. **The base setup.** Someone sits. Overheads off. Key light at **2:00 or 10:00**, above eye level, tilted down; use the large panel. Back light **opposite the key** (if the key is at 2, the back light is at 8), behind and

above, on the hair and shoulders. Look for a catchlight in both eyes and a soft shadow on the far side of the nose.

2. **With fill, without fill.** Capture one still with the fill in (foam core on the side opposite the key, or the second panel at low power). Pull the fill and capture the same frame again. Two stills: fill / no fill.
3. **Around the clock.** Take the fill and the back light out; leave only one light, the one that was the key. Now walk it around the clock and capture a still at every hour: 12, 1, 2, 3, 4, 5, 6 (straight behind), 7, 8, 9, 10, 11. Keep it at the same height and distance each time so the only thing changing is the angle. Notice that the light's name changes as it goes. At 12 through about 4 and 8 through 12 it's a key. At 3 and 9 it's splitting the face in half. At 4, 5, 7, and 8 it's raking from behind: a kicker. At 6 it's straight behind the head: a back light, and the face is a silhouette with a rim. Same lamp, same power; only where it stands decides what it is. Do the whole circle twice: once with the subject **head on** to the camera, once in **profile**. That's up to 24 stills; press the button each time.

## Extra challenges

- **Hard vs. soft.** Same face, two stills: the small panel far back, then the large panel brought in close. Look at the shadow edge and the nose-to-cheek transition. What changed is the apparent size of the source, not its brightness.
- **Fill with a reflector.** Rebuild the base setup, but make the fill with a reflector instead of a panel: hold the white or silver side just out of frame on the side opposite the key and angle it at the face until the shadow side lifts. Capture with and without.
- **Color contrast.** Swap the RGB lights in for two of the three points and set them to different colors: a warm key and a cool back light, or a key in one color and the fill in its opposite (orange and teal, magenta and green). Capture it. Then try one color on the wall behind the subject and white on the face.
- **The ring light.** Put it on the camera axis, right around the lens, and capture. Look at the eyes: one round catchlight, centered, and almost no shadows on the face. That's the look of a lot of what you watch; now you know what made it.

## Notice

Watch the clock series play back on the wall. 12 is flat; 2 and 10 are the portrait angles; 3 and 9 split the face; 4, 5, 7, and 8 rake it from behind; 6 is a silhouette with a rim. Key, kicker, back light: the words are positions, not lamps. The catchlight moves around the eye as the key moves around the clock. That's the vocabulary: when you like a creator's lighting, you can now say where the key was.